

The integration architect's guide to hybrid identity lifecycle with Logic Apps

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**NORDIC
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Identity provisioning · Service desk

04:12

P1

INC-048121

Day 1: New hire anna.lindqvist has no mailbox. UPN mismatch.

Reporter: HR onboarding Assigned: Identity team SLA: 1 h

Every hybrid org has this ticket.

This session is about never getting it again.



whoami

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Modern Security Hero, Not Bad Security

Identity & integration architecture on Azure and Microsoft 365

Microsoft Security MVP

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not**bad**security*

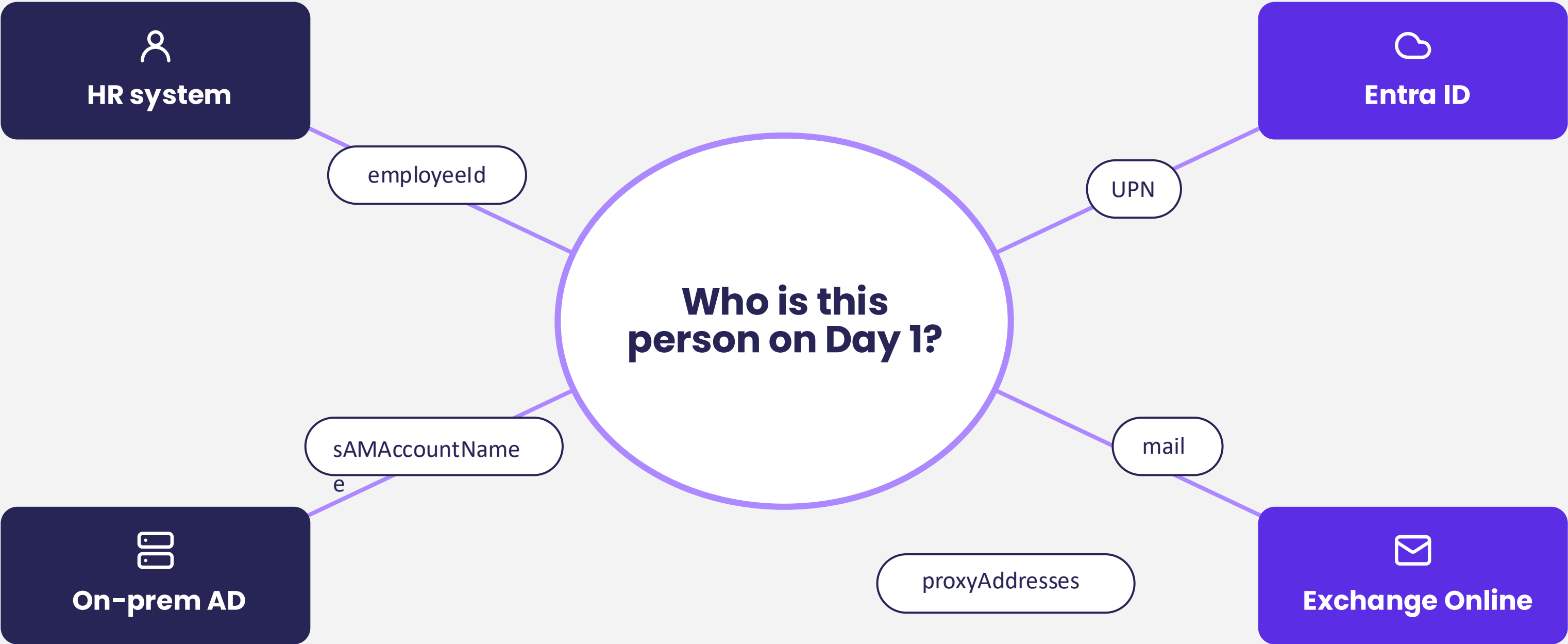
Cloud-first $\neq$ cloud-only.

- Legacy apps still bind to AD
- Kerberos, file servers, printers



Source: Microsoft survey of IT professionals, 2025

Four systems, one question



Where SCIM* stops



Native SCIM / Entra inbound provisioning

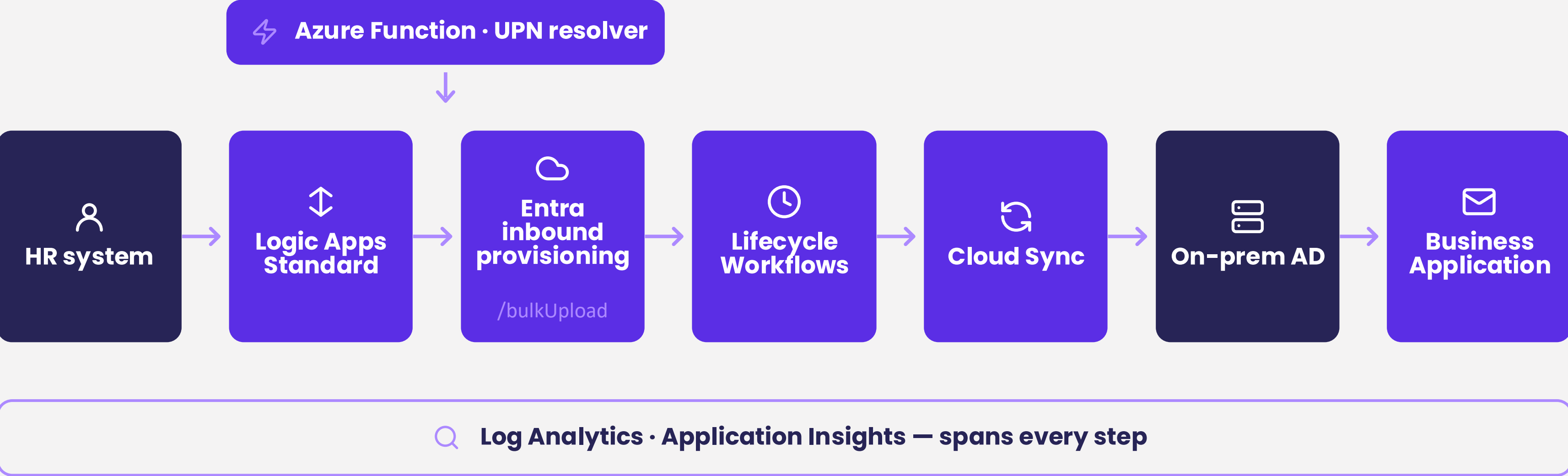
create · update · disable · attribute mapping

Integration work

UPN uniqueness · proxyAddresses · sync timing · cross-system ordering · observability

* SCIM: System for Cross-domain Identity Management, an open standard for provisioning users and groups (RFC 7643/7644)

Reference architecture

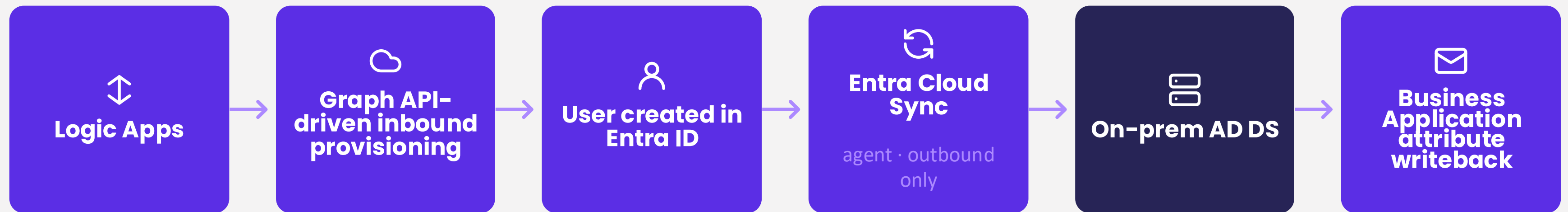


Who is the source of authority?

	Entra ID as SOA Preferred (2026)	AD as SOA
Where user is created	Entra ID (API-driven provisioning)	On-prem AD (provisioning agent)
How AD gets the object	Cloud Sync provisions Entra → AD	Created directly in AD
Sync tool	Cloud Sync agent, outbound only	Connect Sync or Cloud Sync AD → Entra
Business Application attributes	Set in Entra, written back to AD	Set in AD, synced up
On collision	Resolver rejects before create	Sync error, manual fix

PATH A • PREFERRED

Entra → AD with Cloud Sync



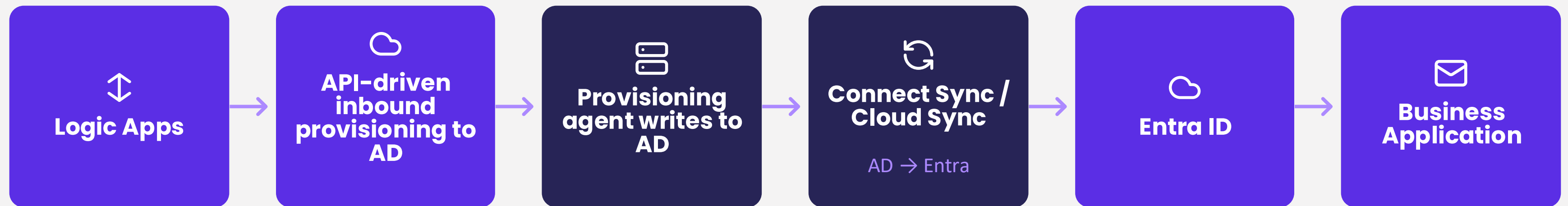
Public preview

Agent ≥ 1.1.2334.0

Schema needs msDS-ExternalDirectoryObjectId

P1 = 2 configs, Governance beyond

API-driven into AD



🕒 **Adds one sync cycle to the Day-1 timeline**

Gotcha #1

Two Annas. One UPN. Non-deterministic = 4 AM.



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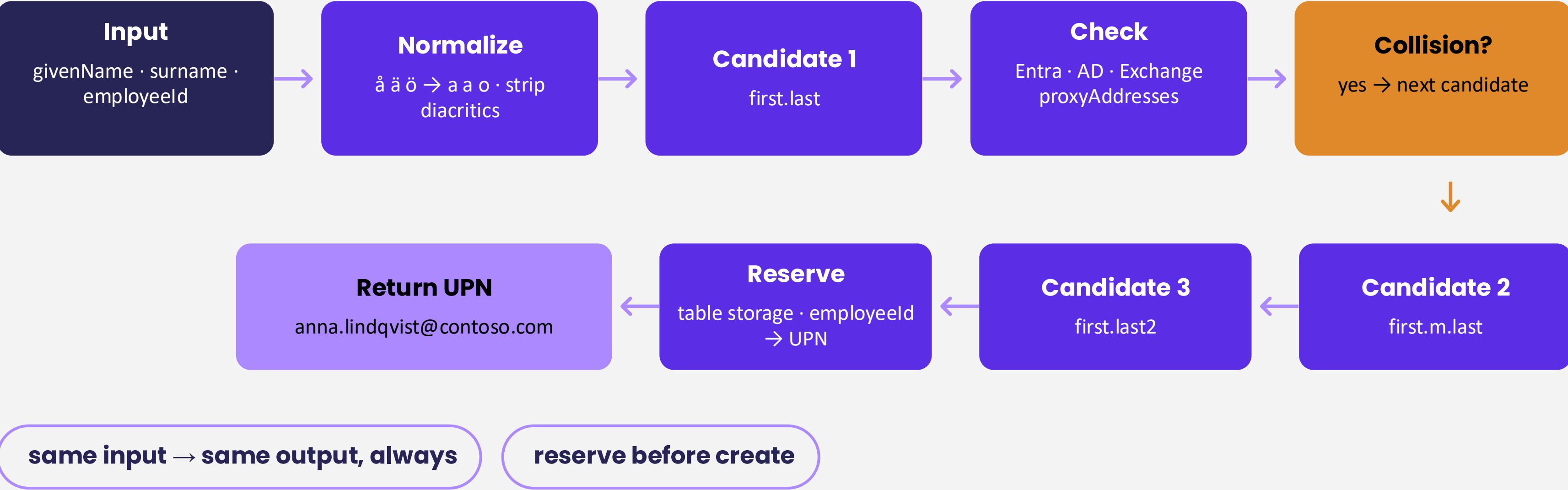


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Deterministic UPN resolver



Calling the resolver from Logic Apps

adapted from working demo

```
{ "type": "Function",
  "inputs": {
    "function": { "id": ".../functions/ResolveUpn" },
    "method": "POST",
    "body": {
      "givenName": "@{triggerBody()?['givenName']}",
      "surname": "@{triggerBody()?['surname']}",
      "employeeId": "@{triggerBody()?['employeeId']}",
      "domain": "lab.lintuala.fi"
    },
  },
  "runAfter": { "Normalize_input": ["Succeeded"] } }
```

Gotcha #2

proxyAddresses decide the mailbox

 **Anna Lindqvist**

user object

proxyAddresses

SMTP: anna.lindqvist@lab.lintuala.fi

primary

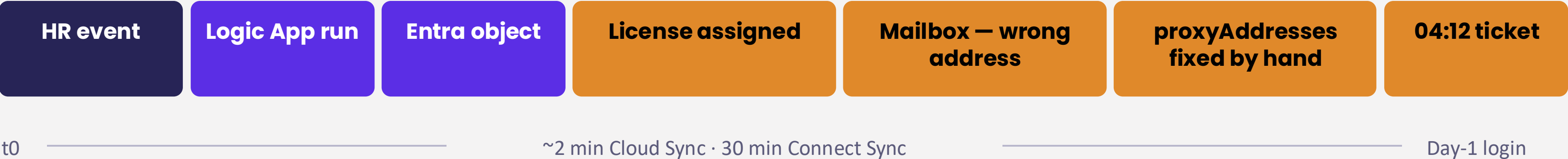
smtp: anna.lindqvist@lablintuala.mail.onmicrosoft.com

smtp: alind@lab.lintuala.fi

- Primary SMTP casing – uppercase means primary
- Missing routing address = mailbox in limbo
- Set before license, not after

Sync timing is architecture

wrong ordering



correct ordering



The "obvious" approach breaks



~~Create in AD and Entra in parallel~~

→ One source, one hop



~~Fixed 30-minute delay step~~

→ Poll for the object



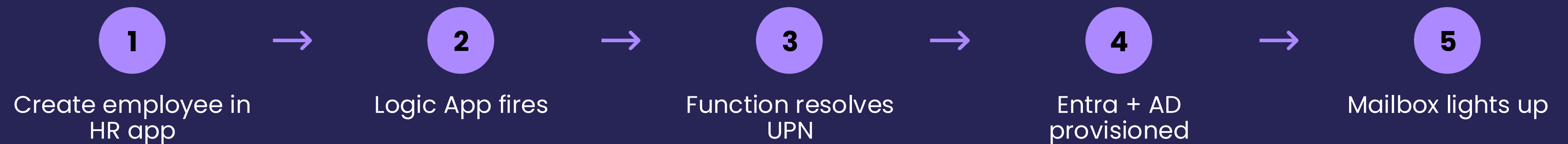
~~Let Exchange pick the address~~

→ proxyAddresses first

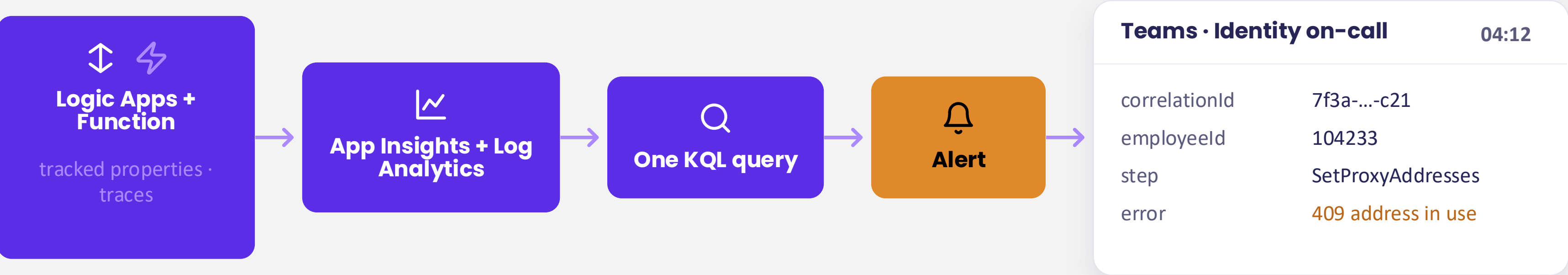
live

Demo: from HR record to lit mailbox

Source of truth: Demo HR application

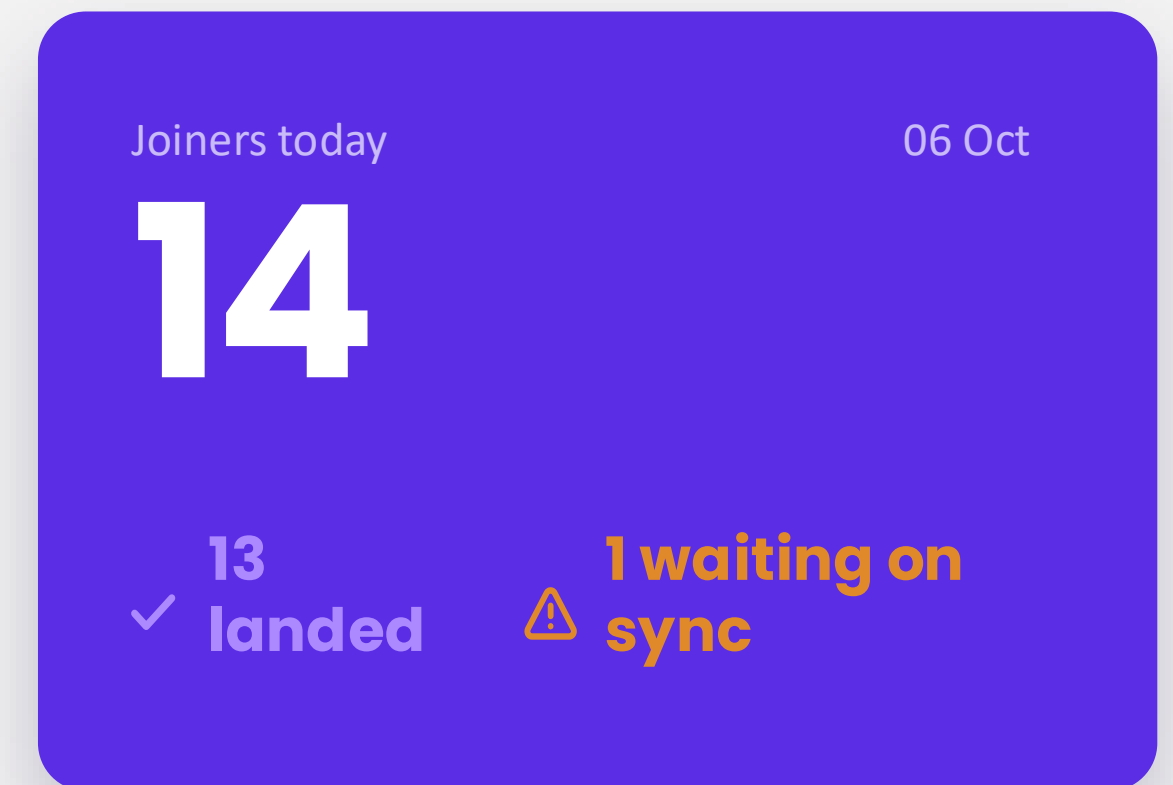


Correlate on employeeld, not on run ID



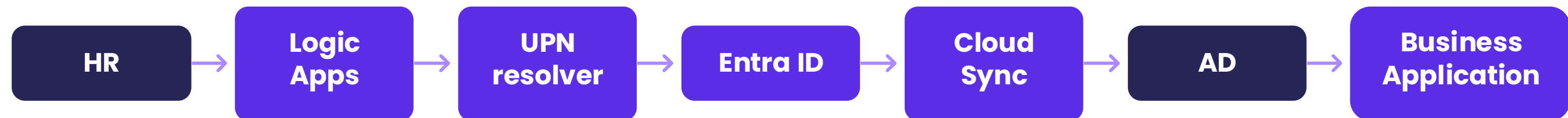
Observability checklist

- ✓ Tracked properties on every action
- ✓ employeeld as correlation key
- ✓ Provisioning logs to Log Analytics
- ✓ Dead-letter for retries
- ✓ One dashboard for joiners



Hybrid identity lifecycle – the pattern

notbadsecurity* · #NIS26



01 Single source of authority

02 Deterministic UPN

03 proxyAddresses before license

04 Design for the sync clock

05 Idempotent every step

06 Observable by employeeId

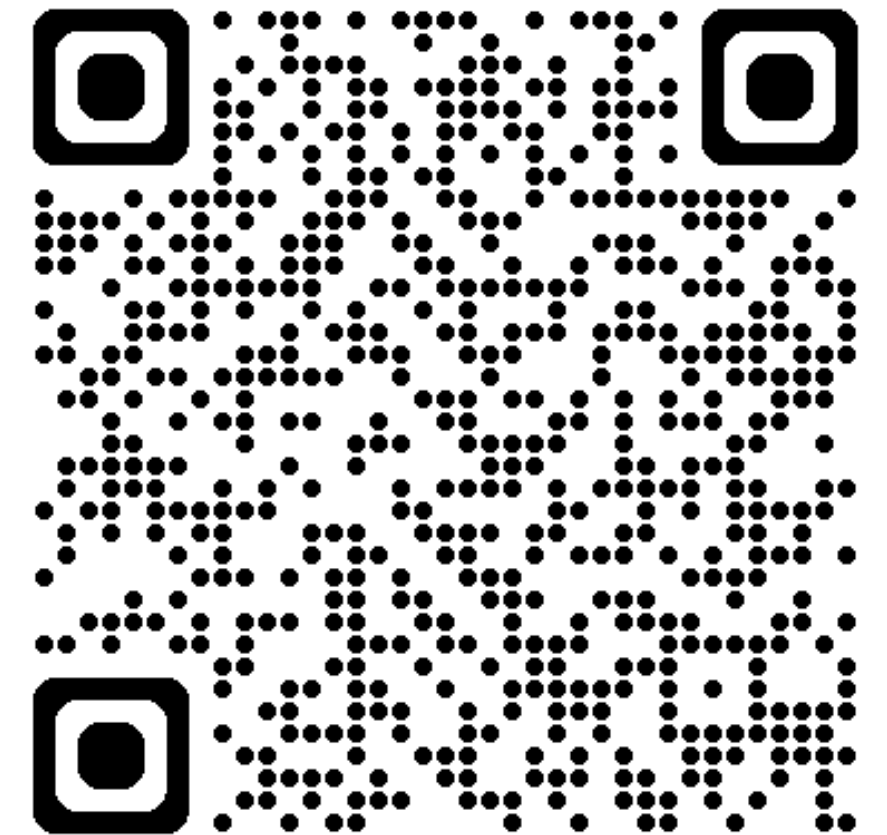
Take it home

☐☐ Proper identity creation process

⚠ Gotcha list with fixes

<> Logic App snippets, KQL queries

github.com/mmaraa/nis26



Thank you

*actually quite hybrid

Questions?

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#NIS26 #azure #identity





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nis
26

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Thank you!